SCIENCE COMMUNICATION CERTIFICATE

The science communication certificate provides an opportunity for students to develop their public communication skills, to interface with students and faculty across disciplinary and science-public divides, and to give students an edge in the job market where successful communication with a multitude of stakeholders is essential. As a discipline, science communication brings together theory and practice to communicate scientific information to the public, with an emphasis on two-way and strategic communication with the public.

This certificate is designed to encourage students who are interested in the intersection of science and society to pursue coursework that provides them with the skills to practice public-facing science and effectively engage the public around complex and sometimes controversial scientific topics. In addition, the certificate is designed to facilitate convergence across disciplines and encourage team-based collaboration at the undergraduate level.

The certificate is open to students of any major, but may be of particular value to students who are planning to pursue science, environmental, health or agricultural communication as a career or students pursuing a career in a science, engineering, math and other technical fields interested in strengthening their communication skills.

Objectives

- Encourage students to engage with community members, other students and faculty across interdisciplinary boundaries, diverse backgrounds, and divergent interests.
- Prepare students to communicate scientific findings and technological advances in a clear and compelling manner while also encouraging inclusive communication that acknowledges others' values and concerns.
- Cultivate students' understanding of the origins and dynamics of science related controversies and conflicts.
- Challenge students to identify and address scientific misinformation, scientific skepticism, and science denial across social media, blogs, and other social and online networks.
- Prepare students to engage in constructive conversations with diverse audiences over contested science, environmental, health, and agricultural topics.

Student Learning Outcomes

Students who complete the proposed science communication certificate will be able to:

- Create, co-produce, and evaluate public-facing science communication
- Understand and address the ethical, social, cultural, and historical factors that influence both the public communication of science and the rise and spread of science-related controversies
- Promote public and cross-disciplinary understanding of scientific information
- Co-produce knowledge with community stakeholders through public-facing projects
- Create effective and appropriate science messages across diverse and emergent media platforms, addressed to diverse audiences
- Critically analyze science messages addressed to public audiences around science
- Identify and address misinformation across social media, blogs, and other social and online networks
- Engage in constructive conversations about contested science, environmental, health, and agricultural topics

To prepare students for the challenge of communicating effectively at the intersection of science and society, students will take a core of 12 credits from the Greenlee School of Journalism and Communication, the English Department, and the Department of Philosophy and Religious Studies and 9 credits of electives from three tracks: science in practice, science and society, and communication in practice.

The certificate requires 21 credits. Students must complete ENGL 2500 before enrolling in the certificate. To successfully complete the certificate, students must submit and receive approval for a final portfolio consisting of a minimum of three applied, public-facing projects developed during or outside of their coursework. A cumulative grade point average of at least 2.0 is required in courses taken for the certificate.

Core Courses (12 credits)

JLMC 2600	Media Controversies in Science and Technology	3
PHIL 2060	Introduction to Logic and Scientific Reasoning	3
ENGL 3120	Communicating Science and Public Engagement	3
JLMC 3470	Science Communication	3

Electives (9 credits)

Students will take one course from each of the three categories of electives: Science in Practice, Science and Society, and Communication in Practice.

- 6 of the 9 credits must be at the 3000+ level.
- Students who engage in an internship or research experience (including those connected to courses) may seek approval from the steering committee to have this experience counted as an elective course. This experience must include a public-facing communication/outreach component, and the steering committee will decide which of the 3 categories of electives this experience would fulfill. Students

who wish to count the internship toward the certificate must track their hours as well as complete a final paper about their experiences.

Science in Practice - select 1 course

BIOL 1730	Environmental Biology	3
BIOL 2510	Biological Processes in the Environment	3
BIOL 3550	Plants and People	3
ENSCI 2500	Environmental Geography	3
ENSCI 3180	Introduction to Ecosystems	3
ENSCI 3600	Environmental Soil Science	3
GEOL 1010	Environmental Geology: Earth in Crisis	3
GEOL 1080	Introduction to Oceanography	3
GEOL 1600	Water Resources of the World	3
GEOL 2010	Geology for Engineers and Environmental Scientists	3
GEOL 2020	History of the Earth	3
GEOL 3240	Energy and the Environment	3
MTEOR 4040	Global Change	3
MTEOR 4060	World Climates	3
NREM 1200	Introduction to Renewable Resources	3
NREM 3800	Field Ecology Teaching	3

Science and Society - select 1 course

EDUC 3470	Nature of Science	3
ENGL 3550	Literature and the Environment	3
HIST 3670	America Eats	3
HIST 3830	European Science, Technology, and Culture	3
HIST 4820	Birth, Death, Medicine, and Disease	3
JLMC 4010	Mass Communication Theory	3
JLMC 4740	Communication Technology and Social Change	3
JLMC 4760	World Communication Systems	3
PHIL 3310	Moral Problems in Medicine	3
PHIL 3340	Environmental Ethics	3
PHIL 3360	Bioethics and Biotechnology	3
PHIL 3430	Ethics of Computing and Artificial Intelligence	3
PHIL 3800	Philosophy of Science	3
PHIL 3890	Philosophy of Psychology and Psychiatry	3
PHIL 4850	Philosophy of Physics	3
POLS 2830	Introduction to Environmental Politics and Policies	3
POLS 3350	Science, Technology, and Public Policy	3
POLS 4430	Energy Policy	3
SOC 3820	Environmental Sociology	3
SOC 4640	Strategies for Community Engagement	3
WGS 3070	Women in Science and Engineering	3
ENVS 3200	Ecofeminism	3
WGS 3800	History of Women in Science, Technology, and Medicine	3
WLC 4840	Technology, Globalization and Culture	3

Communication in Practice - select 1 course

COMST 3270	Persuasion and Social Influence	3
COMST 4120	Health Communication	3
ENGL 3090	Proposal and Report Writing	3
ENGL 3140	Technical Communication	3
ENGL 3550	Literature and the Environment	3
ENGL 4110	Technology, Rhetoric, and Professional Communication	3
ENGL 4770	Seminar in Technical Communication	3
ENGL 4870	Internship in Business, Technical, and Professional Communication	1-3
LING 1200	Language and Technology	3
NREM 3300	Principles of Interpretation	3
PR 2200	Principles of Public Relations	3
PR 3050	Publicity Methods	3
SPCM 2120	Fundamentals of Public Speaking	3
SPCM 3100	Rhetorical Analysis	3